

ID # _____
DATE _____

COMPLETED: YES..... 1
NO..... 5

NEUROLOGICAL EXAM VIDEOTAPED: YES..... 1
NO..... 5

COMMENT: _____

NEUROLOGICAL EXAM

Circle "Normal", "Abnormal", "Other", "Can't execute" or "Missing" for each question.
Always complete "specify" field when noted.

For each question, "Can't Execute" and "Missing" will mean the following:

-**Can't Execute:** subject will not/cannot attempt task secondary to dementia.

-**Missing:** examiner omits task, subject refuses (not secondary to dementia), or subject unable to do task secondary to physical reason.

1. RANGE/EXTENT OF LATERAL GAZE

		<i>Circle</i>
NORMAL	Complete gaze to left/right	1
ABNORMAL	Incomplete left or right gaze (Specify left or right incomplete)	2
	Complete absence of left or right gaze (Specify left or right absence)	3
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/COMMENT:		

2. RANGE/EXTENT OF VERTICAL GAZE

		<i>Circle</i>
NORMAL	Complete up & down gaze	1
ABNORMAL	Incomplete up & down gaze (Specify whether up or down incomplete)	2
	Complete absence of up & down gaze (Specify up or down absence)	3
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/COMMENT:		

3. RAISE EYEBROWS

		<i>Circle</i>
NORMAL	No weakness noted	1
ABNORMAL	Inability to raise eyebrow	2
	Inability to wrinkle forehead on left or right (Specify left or right does not raise as high as other side)	3
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

4. WIDE SMILE- SHOW TEETH

		<i>Circle</i>
NORMAL	No weakness noted	1
ABNORMAL	Flattened nasolabial fold	2
	Inability to raise corner of mouth on left or right (Specify left or right not raised)	3
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

5. PHARYNGEAL MOVEMENTS (subject opens mouth, says "ah")

		<i>Circle</i>
NORMAL	Symmetric palate elevation	1
ABNORMAL	One side not elevated as high (Specify left or right not elevated as high)	2
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

6. UPDRS FACIAL EXPRESSION

		<i>Circle</i>
NORMAL		1
Minimal hypomimia, could be normal "Poker Face"		2
Slight but definitely abnormal diminution of facial expression		3
Moderate hypomimia; lips parted some of the time		4
Masked or fixed facies with severe or complete loss of facial expression; lips parted ¼ inch or more		5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

7. UPDRS Speech

		<i>Circle</i>
NORMAL		1
Slight loss of expression, diction and/or volume		2
Monotone, slurred but understandable; moderately impaired		3
Marked impairment, difficult to understand		4
Unintelligible		5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

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8. FINGER -TO-NOSE TOUCHING

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	Quickly, smooth and accurate	1	1
ABNORMAL	Slow but accurate	2	2
	Dysmetria noted	3	3
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENTS:			

9. UPDRS FINGER- THUMB TAPPING

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	4 taps/second	1	1
ABNORMAL	Mild slowing and/or reduction in amplitude	2	2
	Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.	3	3
	Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement	4	4
	Can barely perform	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

10. UPDRS – Hand Movements –

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	Normal	1	1
ABNORMAL	Mild slowing and/or reduction in amplitude	2	2
	Moderately impaired. Definite and early fatiguing . May have occasional arrests in movement.	3	3
	Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement	4	4
	Can barely perform the task.	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

11. UPDRS – Rapid alternating movements of hands

		Right	Left
		Circle	Circle
NORMAL	At least 3 pats/second and smooth	1	1
ABNORMAL	Mild slowing and/or reduction in amplitude	2	2
	Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.	3	3
	Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement	4	4
	Can barely perform the task.	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

12. HAND PRAXIS TASKS

A). Inter-locking thumbs		Circle
NORMAL	Performs correctly	1
ABNORMAL	Performs incorrectly	2
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

B). Mirrored fingers		Circle
NORMAL	Performs correctly	1
ABNORMAL	Performs incorrectly	2
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

13. UPDRS – Rigidity MUSCLE TONUS (neck)

		Circle
NORMAL	Normal muscle tone, no rigidity	1
ABNORMAL	Slight	2
	Mild to moderate	3
	Marked, but full range of motion easily achieved	4
	Severe, range of motion achieved with difficulty	5
OTHER (SPECIFY)		7

CAN'T EXECUTE	8
MISSING	9
SPECIFY/ COMMENT:	

14. UPDRS – Rigidity MUSCLE TONUS (upper extremity)

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	Normal muscle tone, no rigidity	1	1
ABNORMAL	Slight or detectable only when activated by mirror or other movements	2	2
	Mild to moderate	3	3
	Marked, but full range of motion easily achieved	4	4
	Severe, range of motion achieved with difficulty	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

15. UPDRS – Rigidity MUSCLE TONUS (lower extremity)

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	Normal muscle tone, no rigidity	1	1
ABNORMAL	Slight or detectable only when activated by mirror or other movements	2	2
	Mild to moderate	3	3
	Marked, but full range of motion easily achieved	4	4
	Severe, range of motion achieved with difficulty	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

16. COGWHEEL PHENOMENON

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	No cogwheeling noted	1	1
ABNORMAL	Slight or noticeable rhythmicity throughout	2	2
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/COMMENT:			

17. UPPER EXTREMITY MOTOR STRENGTH

<i>Wrist extension = 1</i>		<i>Triceps pull = 2</i>
		Circle
NORMAL	No strength difference between R and L trials	1
ABNORMAL	Specify weakness (L or R)	2
OTHER (SPECIFY)		7
CAN' T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT		

18. UPDRS – TREMOR AT REST (face, lip, chin)

		Circle
NORMAL	Absent	1
ABNORMAL	Slight and infrequently present	2
	Mild in amplitude and persistent. Or moderate in amplitude, but only intermittently present.	3
	Moderate in amplitude and present most of the time.	4
	Marked in amplitude and present most of the time.	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

19. UPDRS – TREMOR AT REST (hands)

		Right	Left
		Circle	Circle
NORMAL	Absent	1	1
ABNORMAL	Slight and infrequently present	2	2
	Mild in amplitude and persistent. Or moderate in amplitude, but only intermittently present.	3	3
	Moderate in amplitude and present most of the time.	4	4
	Marked in amplitude and present most of the time.	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

20. UPDRS – TREMOR AT REST (feet)

		Right	Left
		Circle	Circle
NORMAL	Absent	1	1
ABNORMAL	Slight and infrequently present	2	2
	Mild in amplitude and persistent. Or moderate in amplitude, but only intermittently present.	3	3
	Moderate in amplitude and present most of the time.	4	4
	Marked in amplitude and present most of the time.	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8

MISSING	9	9
SPECIFY/ COMMENT:		

21. PRONATOR DRIFT (arms outstretched, palms up, eyes closed)

		<i>Circle</i>
NORMAL	Absence of drift	1
ABNORMAL	Spontaneous drift of either/both hands (Specify left, right or both drift)	2
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

22. STRENGTH DIFFERENCE WITH DOWNWARD PRESSURE AND THEN SUDDEN RELEASE BY EXAMINER (arms outstretched, resists examiner's pressure)

		<i>Circle</i>
NORMAL	No strength difference, equal rebound	1
ABNORMAL	Unequal rebound (Specify left or right rebound higher)	2
OTHER (SPECIFY)		7
CAN' T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

23. UPDRS –ACTION OR POSTURAL TREMOR OF HANDS

		<i>Right Circle</i>	<i>Left Circle</i>
NORMAL	Absent	1	1
ABNORMAL	Slight; present with action	2	2
	Moderate in amplitude, present with action	3	3
	Moderate in amplitude with posture holding as well as action	4	4
	Marked in amplitude; interferes with feeding	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

24. DEEP TENDON REFLEXES

	ABSENT	REDUCED	NORMAL	BRISK, NO CLONUS	BRISK, W/ CLONUS	CAN'T EXECUTE	MISSING
<i>Biceps, R</i>	0	1	2	3	4	8	9
<i>Biceps, L</i>	0	1	2	3	4	8	9
<i>BR., R</i>	0	1	2	3	4	8	9
<i>BR., L</i>	0	1	2	3	4	8	9
<i>Knees, R</i>	0	1	2	3	4	8	9

Knees, L	0	1	2	3	4	8	9
AJ, R	0	1	2	3	4	8	9
AJ, L	0	1	2	3	4	8	9
SPECIFY:							

25. ANKLE CLONUS

		Right	Left
		Circle	Circle
NORMAL	Absent	1	1
ABNORMAL	Present	2	2
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

26. UPDRS – LEG AGILITY

		Right	Left
		Circle	Circle
NORMAL	Absent	1	1
ABNORMAL	Mild slowing and/or reduction in amplitude	2	2
	Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.	3	3
	Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement	4	4
	Can barely perform the task.	5	5
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/ COMMENT:			

27. PROPRIOCEPTION

	Right	Left
	Circle	Circle
PRESENT	1	1

ABSENT	2	2
CAN'T EXECUTE	8	8
MISSING	9	9
SPECIFY/COMMENT:		

28. VIBRATING SENSATION

	<i>Right</i>	<i>Left</i>
	<i>Circle</i>	<i>Circle</i>
PRESENT	1	1
ABSENT	2	2
CAN'T EXECUTE	8	8
MISSING	9	9
SPECIFY/COMMENT		

29. PLANTAR RESPONSE

		<i>Right</i>	<i>Left</i>
		<i>Circle</i>	<i>Circle</i>
NORMAL	Plantar flexion of great toe	1	1
ABNORMAL	Extension of great toe	2	2
	No reflex present	3	3
OTHER (SPECIFY)		7	7
CAN'T EXECUTE		8	8
MISSING		9	9
SPECIFY/COMMENT			

30. UPDRS – ARISING FROM CHAIR

		<i>Circle</i>
NORMAL		1
ABNORMAL	Slow; or may need more than one attempt	2
	Pushes self up from arms of seat.	3
	Tend to fall back and may have to try more than one time, but can get up without help.	4
	Unable to arise without help.	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

31. ROMBERG'S SIGN (stand with feet together for 10-15 seconds)

		<i>Circle</i>
NORMAL	Normally still or slight weaving	1
ABNORMAL	Falls to one side with eyes closed (Specify falls to left or right)	2
	Falls to one side with eyes open (Specify falls to left or right)	3
	Needs widened base to stay in one place	4

OTHER (SPECIFY)	7
CAN'T EXECUTE	8
MISSING	9
SPECIFY/ COMMENT:	

32. UPDRS – POSTURAL STABILITY

52. GDRS - POSTURAL STABILITY		Circle
NORMAL		1
ABNORMAL	Retropulsion, but recovers unaided.	2
	Absence of postural response; would fall if not caught by examiner	3
	Very unstable, tends to lose balance spontaneously	4
	Unable to stand without assistance	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

33. UPDRS GAIT – and also time gait task

55. OTHER GAIT and also time gait task		
A). Walking down a hall at least 10 paces		Circle
NORMAL	Normal gait	1
ABNORMAL	Walks slowly, may shuffle with short steps, but no festination (hastening steps) or propulsion	2
	Walks with difficulty, but requires little or no assistance; may have some festination, short steps, or propulsion	3
	Severe disturbance of gait, requiring assistance	4
	Cannot walk at all, even with assistance	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

B). Pivot while turning		Circle
NORMAL	Pivots on narrow base	1
ABNORMAL	Hesitates	2
	Widens base or moves feet	3
	Turns slowly or awkwardly	4
OTHER (SPECIFY)		7
CAN' T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

<i>C). Timed gait task</i>	
Trial 1	— · — Seconds

Trial 2		--·-- Seconds
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		
Type of floor surface	Linoleum/tile/wood	1
	Low-pile carpet	2
	High-pile carpet	3
	Concrete	4
	Other (specify)	7
	DK	98
SPECIFY/ COMMENT:		
Type of aid used	None	1
	Walking stick or cane	2
	Crutches	3
	Walking frame	4
	Other (Specify)	7
	DK	98
SPECIFY/COMMENT:		

34. UPDRS –BODY BRADYKINESIA AND HYPOKINESIA (Combining slowness, hesitancy, decreased arm swing, small amplitude, and poverty of movement in general).

		Circle
NORMAL	None	1
ABNORMAL	Minimal slowness, giving movement a deliberate character; could be normal for some persons. Possibly reduced amplitude	2
	Mild degree of slowness and poverty of movement which is definitely abnormal. Alternatively, some reduced amplitude.	3
	Moderate slowness, poverty or small amplitude of movement	4
	Marked slowness, poverty or small amplitude of movement	5
OTHER (SPECIFY)		7
CAN' T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT		

35 MYOCLONUS

		Circle
NORMAL	Absent	1
ABNORMAL	Mild myoclonus	2
	Occasional myoclonus	3
	Frequent myoclonus	4
	Severe myoclonus	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9

SPECIFY/ COMMENT:

36. UPDRS - POSTURE

		<i>Circle</i>
NORMAL	Normal, erect	1
ABNORMAL	Not quite erect, slightly stooped; could be normal for older person	2
	Moderately stooped posture, definitely abnormal; can be slightly leaning to one side (Specify if leaning to right, left or neither)	3
	Severely stooped posture with kyphosis; can be moderately leaning to one side (Specify if leaning to right, left or neither)	4
	Marked flexion with extreme abnormality of posture	5
OTHER (SPECIFY)		7
CAN'T EXECUTE		8
MISSING		9
SPECIFY/ COMMENT:		

37. PRAXIS TASKS

PRETEND TO	NORMAL PERFORMS CORRECTLY	ABNORMAL (SPECIFY)	OTHER (SPECIFY)	CAN'T EXECUTE	MISSING
<i>Comb your hair</i>	1	2	7	8	9
<i>Hammer a nail</i>	1	2	7	8	9
<i>Brush your teeth</i>	1	2	7	8	9
SPECIFY/COMMENT:					

COMMENTS

This neurological examination was modified from similar instruments for assessment of Huntington's disease(Folstein,SE et al., Neurobehavioral Toxicology and Teratology,1983,5:605-609), the CERAD exam for AD (Morris,JC et al., Neurology, 1989, 39:1159-1165) and the UPDRS. (Fahn, S. Elton, R.L., UPDRS Development Committee; The Unified Parkinson's Disease Rating Scale. In Fahn S, Marsden CD, Calne DB, Goldstein M, eds. Recent developments in Parkinsons's disease, 1987: 153-163, 293-304.)